BUDAPESTI CORVINUS EGYETEM

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Operational Risk

Risk Management and Financial Institutions

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Operational risk definition

Operational risk:

- internal processes,
- systems,
- people,
- or risk of loss due to some external events

The definition includes legal risk:

- Risk of current or future loss (having negative effect on capital or results), occurred by the company’s non compliance to legal regulations (laws and other regulations), agreements or any other specified practices, ethical standards or violation of any previous regulations.
- Regulation definition does not cover strategic and reputational risks, however - based on supervisory approach - reputational risks is part of operational risks.

Capital calculation methods

BIA
15%* authoritative indicator

No precondition

STA
12%/15%/18% * authoritative indicator

- Branch classification
- Risk management
- Data collecting
- Report/ handling
- Documented processes
- Validation- internal and external

AMA
Internal model, 99,9% percentile

- Data (Internal data– 5 years – External data)
- Modelling – scenario analysis
- Validation- internal/external
- Taking into account insurance

ASA – Alternative Standard Method

From 2025 SMA method: Gross income based, corrected by losses

Basic Indicator Approach (BIA)

- Under this method, the regulatory capital requirement (K_{BIA}) can be expressed as a fixed percentage (α) of the bank's average annual gross income (GI) over the previous three years.

- $K_{BIA} = GI * \alpha$, where α is 15%.

- Gross income (GI) is the sum of net interest income and non-interest income, including provisions (including contingent interest), but does not include realized and unrealized foreign exchange gains/losses on the sale of banking book securities or extraordinary income items and insurance income.

Standardised Approach (STA)

- In the standard method, the capital requirement to be formed (K_{STA}) is calculated by adopting the elements and method of calculation of the basic indicator method. The capital requirement for the operational risk of the given area is determined as a fixed percentage (β_{1-8}) of the average annual gross income (GI_{1-8}) by business area.

- $K_{STA} = \Sigma\ (GI_{1-8} * \beta_{1-8})$

Alternative Standardised Approach (ASA)

- Its purpose is to eliminate the double penalty for credit risk. In the case of the retail and commercial banking business, the indicator (LA) for total loans and advances, multiplied by a fixed coefficient ($m = 0.035$), replaces gross income.

- $GI = LA * m$

Advanced Measurement Approach – Qualitative requirements

- A well-documented operational risk management unit responsible for the development and application of the operational risk management system, policy, methodology and strategy.
- Regular reports for senior management
- It is an integral part of daily risk management processes
- The bank's risk management system shall operate in accordance with internal policies, policies and procedures, which shall include non-compliance.
- Internal and external / supervisory control

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Advanced Measurement Approach – Quantitative requirements

- Model
- Loss data collection
- Risk and control self-assessment
- Scenario analysis
- Key risk indicators

AMA model

- Risk financing - taking insurance into account
- Separate modeling of frequency and impact of losses
- Loss frequency:
  - Discrete distribution (binomial, negative binomial, poisson)
- Loss size:
  - Exponential, lognormal, pareto

SMA method

1. BI (Business Indicator – gross income)
2. BI component (BIC)
3. LC (loss component)
4. ILM (Internal loss multiplier)
5. Calculation of capital

$$\textit{Loss component} = 15 * \text{Average loss amount (10 years)}$$

$$ILM = \ln\left(\exp(1) - 1 + \left(\frac{LC}{BIC}\right)^{0{,}8}\right)$$

$$\text{ORC} = \text{BIC} * \text{ILM}$$

BI component:

BI (mrd eur)	BI component (mrd eur)
0 – 1	12%*BI
1 – 30	0,12 + 15%*(BI – 1)
>30	4,47 + 18%*(BI – 30)

Thank you for your attention!

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